Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element H Initial Template

Element H: Prototype Testing and Data Collection Plan

Testing Details *(H1, H2, H3, H4)*

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
3	Grip tightly onto the string	1. Attach the prototype to a hoodie string 2. Hold the hoodie by the sleeves 3. Wring the hoodie around several times (throw it around, shake it up, toss it around)	Test cases 1 **Trials 3**	Fall off<stay on<gets stuck	Fails: falls off immediately after some wrining Pass: stays on and does not get stuck to any parts of the hoodie Goal: does not get stuck but also not fall off of the string
4	Be sturdy	1. Take the prototype and put it on the floor 2. Step directly onto the prototype ~ 120-130lbs	Test cases 1 Trials: 3	breaks<reading	Fails: the prototype immediately breaks upon foot contact Pass: the object is still working and not broken after impact Goal: nothing is bent out of shape and a new prototype does not need to be made
5	Keep the string	1. Attach the prototype onto the	Test cases 1	String gets	Fails: prototype

Element H continues on the next page →

	from not getting lost	string on one side 2. Take the other non attached side and attach to a 20lb weight 3. Lift hoodie from the ground keeping the weight a little suspended in the air 4. See if the string stays in place	Trials 3	lost<stays out	slips and falls off, allowing the string to be pulled completely through Pass: the prototype is anchored tightly and the string does not get lost Goal: Prototype stays out keeping the string out
6	Effortlessly put on	1. Get a timer 2. Time the time it takes to put the prototype on 3. Get the average	Test cases 1 Trials 3	0<reading<60 secs	Fails: the prototype takes more than one minte to put on Pass: prototype takes less than 60 secs to be put on Goal: prototype only takes about 10 secs to be put on and take off

Element H continues on the next page →

Feedback From Field Experts *(H5)*

Mr. Sid - what about the noise it will make in the dryer

- Think about the way to prevent that or test the noise
- How will you know if it will melt in the dryer
- Maybe use heat or a hair dryer to test that

Mr. Steinhorn -

- How well do you think it will last
- Will you need to adjust the device every single time you use it

Revised Testing Details That Incorporate Field Expert Feedback *(H5)*

7	Effortlessly stays on (Do this after the shake test)	1. Inspect the device on the string and see if it has gotten loose	Test cases 1 Trials 1	loose<tight	Fails: the prototype is loose and needs to be tightened Pass: prototype is not loose but is starting to slip a little bit Goal: prototype still securely on without any slipping and will look like it will stay that way

Element H: Prototype Testing and Data Collection Plan						
	5	4	3	2	1	0
1. Addressing Requirements. The final prototype Testing Plan addresses ______________ of the *high priority* design requirements.	*all or nearly all*	*many*	*some*	*a few*	*one*	*none - or is missing altogether*
2. Conduct of Tests. The Testing Plan ______________ in describing the conduct of tests (through physical and/or mathematical modeling).	*is effective*	*is generally effective*	*is only adequate*	*is partially effective*	*is at least minimally effective*	*is void*
3. Quantity of Tests Planned. The Testing Plan defines ___________________ of *all* design requirements based on the instructional context.	sufficient quantity to feasibly test *100%*	sufficient quantity to feasibly test *80%*	sufficient quantity to feasibly test *60%*	sufficient quantity to feasibly test *40%*	*sufficient quantity to feasibly test 20%*	*no feasible test*

4. Logic of the Plan. The Testing Plan provides _______________________.	a logical and *well -developed* explanation	*a logical and generally developed* explanation	*a generally logical and adequately developed* explanation	*a somewhat logical and partially-developed* explanation	*at least a generally logical and/or partially -developed explanation*	*no explanation*
5. Use of Field Experts. The Testing Plan is confirmed by _____________ .	*more than two* field experts with feedback provided and incorporated	*two* field experts with feedback provided and incorporated	one field expert with feedback provid*ed and incorporated*	one field expert *with feedback provided*	*one field expert but no response was documented*	*Field expertise is not mentioned or addressed and field experts are not consulted.*